

Approximating the Score Function using Optimal Transport

Curves and Surfaces 2026

Quentin Giton

Laboratoire de Mathématiques d'Orsay

June 11th

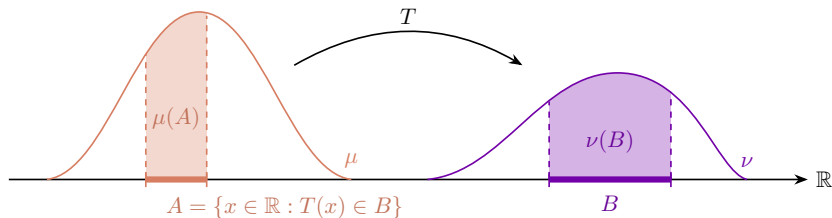
 (Badly and Slowly) Approximating the Score Function
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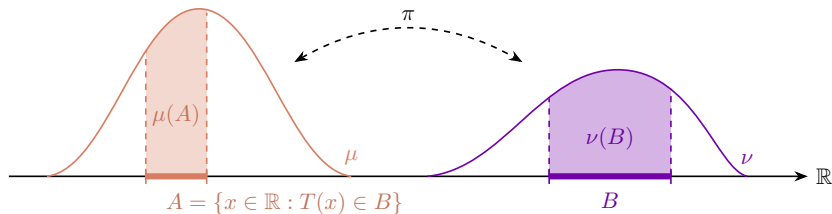
Quick reminders on Optimal Transport 1/3



- ▶ Given ground cost $c(x, y)$
- ▶ Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$
- ▶ Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x)) \mu(dx)$$

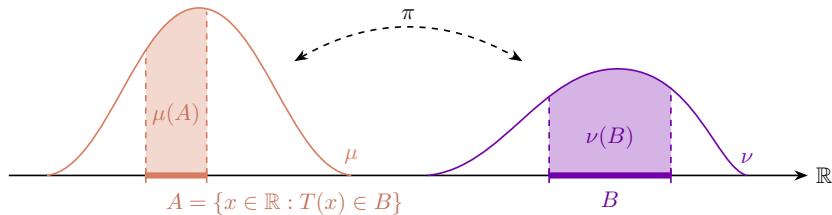
Quick reminders on Optimal Transport 1/3



- ▶ Given ground cost $c(x, y)$
- ▶ Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$
Among all **transport plans** π with marginals μ and ν
- ▶ Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x)) \mu(dx) \longrightarrow \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \pi(dx, dy)$$

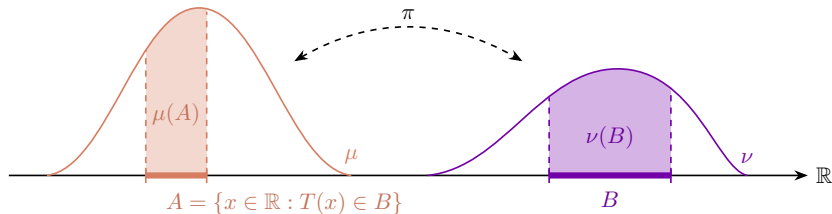
Quick reminders on Optimal Transport 2/3



► p -Wasserstein distance:

$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p \gamma(dx, dy)$$

Quick reminders on Optimal Transport 2/3



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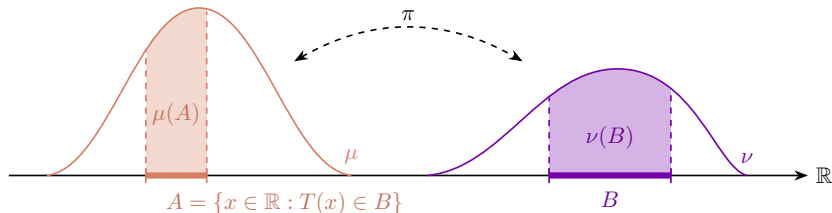
$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p \gamma(dx, dy)$$

- Kantorovich duality

$$\inf_{\pi \in \Pi(\mu, \nu)} \int c(x, y) \gamma(dx, dy) = \sup_{\phi} \int \phi(x) \mu(dx) + \int \phi^c(y) \nu(dy),$$

where $\phi^c(y) := \inf_x c(x, y) - \phi(x)$.

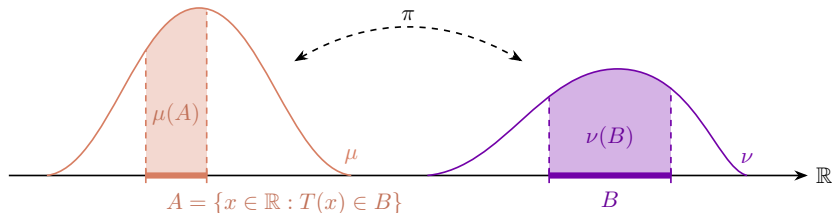
Quick reminders on Optimal Transport 3/3



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Quick reminders on Optimal Transport 3/3



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► If an optimal transport map T^* exists, there also exists an optimal Kantorovich potential ϕ^* and

$$y = T^*(x) = \nabla \left(\frac{\|\cdot\|^2}{2} - \phi^* \right) (x) = x - \nabla \phi^*(x),$$

hence $-\nabla \phi^*(x) = T^*(x) - x = y - x$ is the **displacement** of the particle x .

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$d\vec{X}_t = -\vec{X}_t dt + \sqrt{2}d\omega_t, \quad \vec{X}_0 \sim \mu, \quad t \in [0, T].$$

Let $\vec{\rho}_t := \text{Law}(\vec{X}_t)$ and $\gamma \propto e^{-\|\cdot\|^2/2}$. Then $\vec{\rho}_t$ solves the Fokker–Planck equation

$$\partial_t \vec{\rho}_t(x) = \Delta \vec{\rho}_t(x) + \text{div}(\vec{\rho}_t(x)x).$$

whose stationary solution is γ . The process $(\vec{X}_t)_t$ is a **diffusive** process.

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- **Backward process:** $\tilde{X}_s := \vec{X}_{T-s}$ for $s \in [0, T]$. If $(\vec{X}_t)_{t \in [0, T]}$ is an OU process, then its time reversal $(\tilde{X}_s)_{s \in [0, T]}$ solves

$$d\tilde{X}_s = \left[\tilde{X}_s + 2 \underbrace{\nabla \ln \left(\vec{\rho}_{T-s}(\tilde{X}_s) \right)}_{\text{The score function}} \right] ds + \sqrt{2}d\omega_s, \quad \tilde{X}_0 \sim \vec{\rho}_T.$$

Using OT to learn the score 1/2

- The Fokker–Planck equation is the 2-Wasserstein Gradient Flow (WGF) of the relative entropy $\mathcal{H}(\rho|\gamma) := \int V\rho + \int \rho \ln(\rho)$. That is:

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \operatorname{div}(\rho_t(x)x) = -\nabla \cdot (\rho_t(x)v_t(x)),$$

where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

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- Given a time-step $\tau > 0$, such a gradient flow can be discretised in time using the **JKO scheme**

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg \min \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

Using OT to learn the score 2/2

► Solving

$$\inf_{\rho} \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2$$

amounts to finding the optimal transport map between

$$\rho_n^\tau \quad \text{and} \quad \rho_{n+1}^\tau \propto \exp\left(-\phi^{c/2\tau}\right) d\gamma,$$

where ϕ solves

$$\inf_{\phi} \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} d\gamma\right) - \int \phi d\rho_n^\tau.$$

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► The optimal potential ϕ^* satisfies

$$-\nabla\phi^*(x) = \nabla(\phi^*)^{c/2\tau}(y) = \frac{y - x}{\tau}.$$

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- Solving

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$$\inf_{\phi} \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} d\gamma\right) - \int \phi d\rho_n^{\tau}.$$

- The optimal potential $\phi^{\star}$ satisfies

$$-\nabla\phi^{\star}(x) = \nabla(\phi^{\star})^{c/2\tau}(y) = \frac{y - x}{\tau}.$$

- The displacement of the particle x being $v_t(x)$, we obtain

$$-\nabla\phi^{\star}(x) = v_t(x) = -x - \nabla\ln(\rho_{n+1}^{\tau}(x)) \iff \boxed{\nabla\ln(\rho_{n+1}^{\tau}(x)) = \nabla\phi^{\star}(x) - x}.$$

General contributions

Proposition (G.)

The functional $\tilde{J}(\phi) := \ln \left(\int e^{-\phi^{c/2\tau}} d\gamma \right) - \int \phi d\rho_n^\tau$ is convex.

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Theorem (G.)

Let

$$J(\phi) := \int e^{-\phi^{c/2\tau}} d\gamma - \int \phi d\rho_n^\tau.$$

Then,

$$\arg \min_{\phi} J(\phi) \subseteq \arg \min_{\phi} \tilde{J}(\phi) \quad \text{and} \quad \arg \min_{\int e^{-\phi^{c/2\tau}} d\gamma = 1} \tilde{J}(\phi) = \arg \min_{\phi} J(\psi).$$

Stochastic gradient descent 1/2

$$\blacktriangleright J(\phi) = \mathbb{E}_{\mathbf{y} \sim \gamma} \left[\exp \left(-\phi^{c/2\tau}(\mathbf{y}) \right) - \int \phi(\mathbf{x}) \rho_n^\tau(\mathrm{d}\mathbf{x}) \right]$$

Stochastic gradient descent 1/2

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- ▶ Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N} \sum_{i=1}^N \delta_{\hat{\mathbf{x}}_i^{(n)}}$$

Stochastic gradient descent 1/2

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- ▶ Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N} \sum_{i=1}^N \delta_{\hat{\mathbf{x}}_i^{(n)}}$$

- ▶ $G(\theta) = \mathbb{E}_{\mathbf{y} \sim \gamma} \left[\max_{1 \leq i \leq N} \left\{ \exp \left(\theta_i - \frac{c(\hat{\mathbf{x}}_i^{(n)}, \mathbf{y})}{2\tau} \right) \right\} - \frac{1}{N} \sum_{i=1}^N \theta_i \right]$

Stochastic gradient descent 1/2

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- ▶ Projected SGD algorithm ($c_\tau(\mathbf{x}, \mathbf{y}) := \frac{c(\mathbf{x}, \mathbf{y})}{2\tau}$)

$$\left\{ \begin{array}{l} \text{pick } \mathbf{y} \sim \gamma \\ \text{let } i := \arg \min_j c_\tau(\hat{\mathbf{x}}_j^{(n)}, \mathbf{y}) - \theta_j \\ \text{define } \mathbf{e}_i \text{ as the } i\text{-th canonical basis vector of } \mathbb{R}^N \\ \theta^{(k+1)} := \text{proj}_{B(0, R_{\text{proj}})} \left(\theta^{(k)} - \alpha_k e^{\theta_i^{(k)} - c_\tau(\hat{\mathbf{x}}_j^{(n)}, \mathbf{y})} \mathbf{e}_i + \frac{\alpha_k}{N} \mathbf{1} \right) \end{array} \right.$$

Stochastic gradient descent 2/2

Theorem (G .)

Let $(\theta^{(k)})_{k=0}^K$ be the iterates obtained with step-sizes $\alpha_k = \frac{R_{\text{proj}}}{B\sqrt{k}}$ starting from some initial $\theta^{(0)} \in B(0, R_{\text{proj}})$. Define $\bar{\theta}_K$ to be the arithmetic mean of all iterates up to time K . Then, it holds that

$$0 \leq \mathbb{E}[G(\bar{\theta}_K)] - G(\theta^*) \leq BR_{\text{proj}} \left(\frac{2}{K} + \frac{1}{\sqrt{K}} \right) = \mathcal{O}(K^{-\frac{1}{2}}),$$

where θ^* denotes the unique minimiser of G .

► Recall that